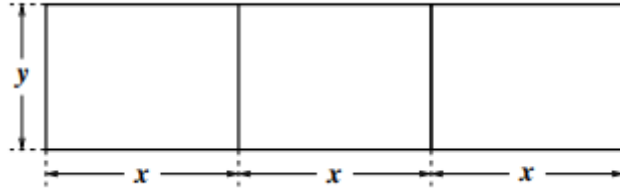


4.7 Optimization Worksheet

1. A farmer wishes to fence off three identical adjoining rectangular pens as in the diagram shown, but she only has 600ft of fencing available. Determine the values for x and y which will maximize the total area enclosed by these three pens.



2. A farmer wishes to enclose a rectangular pen with area 100ft^2 next to a road. The fence along the road is to be reinforced and costs \$34 per foot. Fencing that costs \$16 per foot can be used for the other three sides. What dimensions for the pen will minimize the cost to the farmer? What is the minimum cost?

3. A rectangular storage container with an open top is to have a volume of $10m^3$. The length of its base is twice the width. Material for the base costs \$10 per square meter. Material for the sides costs \$6 per square meter. Find the cost of materials for the cheapest such container?

4. Suppose $y = \frac{8}{x^6}$ for $x > 0$. Determine the x -coordinate for the point on this curve which is closest to the origin.

5. The top and bottom margins of a poster are each 6cm and the side margins are 4cm. If the area of the printed material on the poster is fixed at 384cm^2 , find the dimensions of the poster with the smallest area.

6. Find the area of the largest rectangle that can be inscribed in a semicircle of radius 3. (Hint: sketch your semi-circle using the top half of the circle $x^2 + y^2 = (3)^2$)

7. A man launches his boat from the bank of a straight river which is 3 km wide, and wants to reach a point on the opposite bank 8 km downstream as quickly as possible. The man could row straight across the river and then run down the bank 8 km to his destination, he could row all the way to his destination, or he could row to some other point on the opposite bank and then run to his destination. If he can row 6 km/h and run 8 km/h , where should he land on the opposite bank to reach his destination as soon as possible? What is the minimum time? (assume the speed of the water is negligible)

8. An open box is to be made from a 16-inch by 30-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume?